

AI-Based CCTV People Tracking & Dwell-Time Analytics System

Autonomous Multi-Camera Surveillance, Deep Spatio-Temporal Re-Identification & Footfall Auditing

A comprehensive technical report detailing the design, algorithmic methodology, and experimental validation of an enterprise-grade surveillance system. The project integrates deep object detection, containment deduplication, motion modeling, multi-exemplar deep Re-Identification (ReID), and automated transactional journey logging to deliver reliable pedestrian tracking across public spaces.

CORE OBJECT DETECTOR

Ultralytics YOLO11m (COCO Pretrained, FP32/FP16)

APPEARANCE REID MODEL

MobileNetV3 (576-D) + 3-Zone CIELAB Color Histograms

PERSISTENCE ARCHITECTURE

ACID SQLite Database + Visual Crop Thumbnail Cache

STATE TRACKING ENGINE

BoT-SORT + Camera Motion Compensation (CMC)

VIEWPOINT MEMORY BANK

Multi-Exemplar Gallery (Top-5 Viewpoint Diversity)

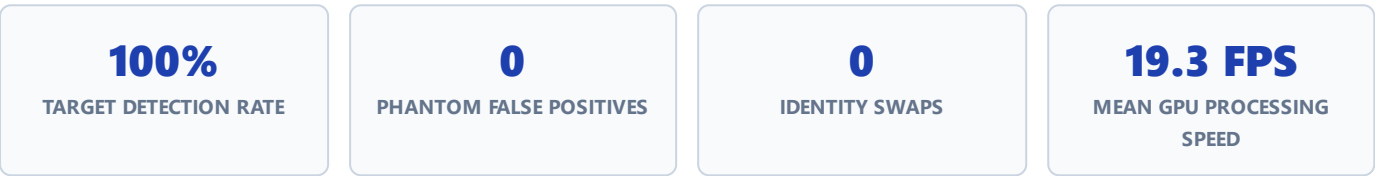
MONITORING SURFACES

Synchronized OpenCV Dual-Panel HUD & Streamlit Web Portal

1. Executive Summary

Modern commercial infrastructure, transit terminals, and retail establishments require accurate real-time intelligence regarding visitor volume, dwell durations, and trajectory pathways. Traditional surveillance infrastructure operates primarily as passive video storage, requiring human review to extract actionable data. Commercial tracking software frequently degrades under dense occlusions, perspective shifts, and background activity, leading to identity fragmentation, duplicate counting, and phantom detections caused by store shelves, glass reflections, and counter structures.

This project engineered an autonomous, real-time edge surveillance pipeline that solves these fundamental challenges. The system detects pedestrians across challenging perspectives, maintains persistent **Global IDs** (e.g., `Person-001`) across turns and camera transitions, logs exact millisecond-precision entry and exit timestamps to a relational database, and provides dual analytical surfaces: an edge OpenCV HUD and a browser-based management portal.



2. Problem Statement & System Objectives

2.1 Core Computer Vision Challenges Addressed

- **Scale & Perspective Variance:** Foreground individuals occupy thousands of pixels, whereas background corridor pedestrians occupy fewer than 50 pixels in height, requiring high detector sensitivity without inducing hallucinated bounding boxes.
- **Counter Occlusions & Ghost Tracks:** Seated staff behind checkout counters or customers walking in close proximity create overlapping bounding boxes that standard Non-Maximum Suppression (NMS) fails to clean.
- **Cross-Camera Re-Identification:** Viewpoint orientation changes and brief exits degrade standard Euclidean distance trackers, causing split identity fragments.
- **Auditable Footfall Accounting:** Security and facility managers require verifiable entry/exit timestamps, dwell durations, and visual image evidence rather than opaque counter values.

2.2 Key Engineering Objectives

- Integrate **YOLO11m** for high-recall pedestrian detection with height and area thresholds.
- Eliminate double-counting on cashier partitions using **Intersection-over-Smaller (IoS)** containment deduplication.
- Formulate a **Spatio-Temporal Deep ReID** model fusing 576-D MobileNetV3 convolutional features with 3-Zone CIELAB color distributions.
- Maintain a **Multi-Exemplar Memory Gallery** (up to 5 viewpoints per person) to bridge occlusions and orientation shifts.
- Persist visitor journeys into an ACID-compliant **SQLite** database with high-resolution thumbnail crops.
- Deploy dual monitoring interfaces: an **OpenCV Synchronized HUD** for edge operators and a **Streamlit Web Administrator Portal** for analytics.

3. System Architecture & End-to-End Workflow

The system is organized into a modular, decoupled pipeline ensuring low processing latency and predictable memory management across all computational modules:

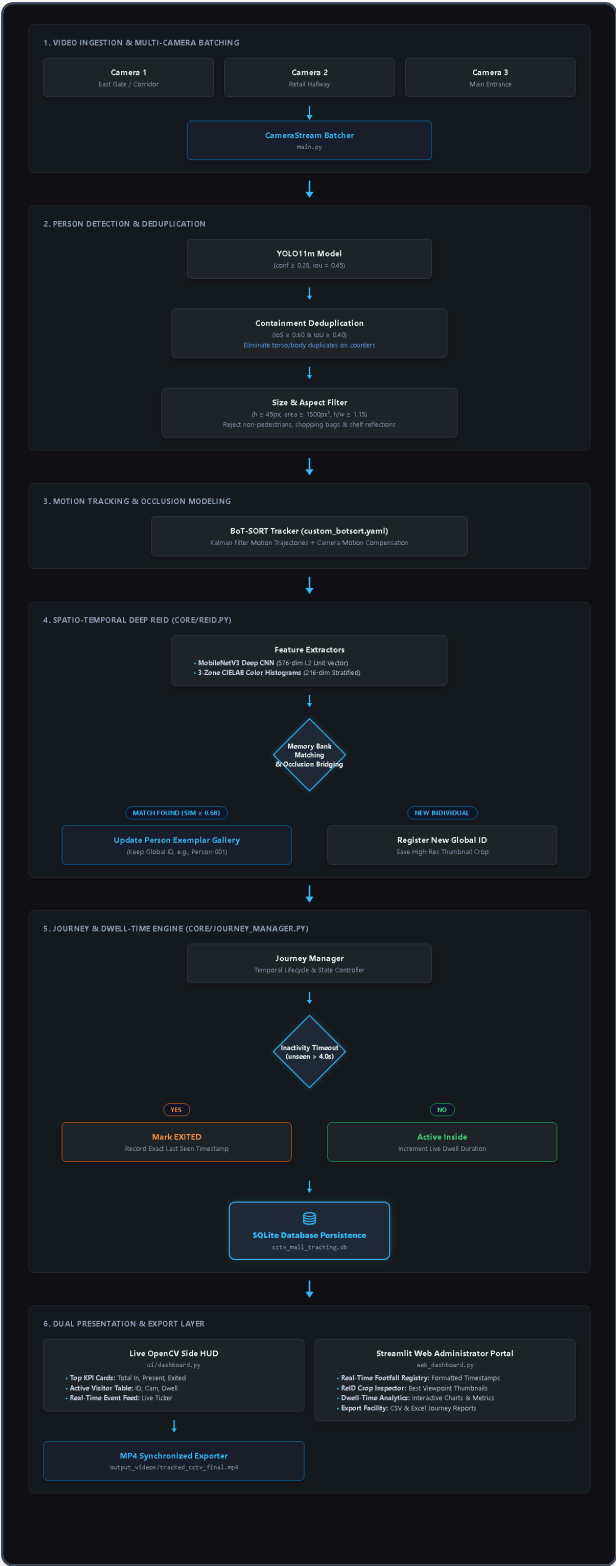


Figure 1: End-to-End System Architecture & Multi-Stage Deep Learning Pipeline Workflow.

4. Algorithmic Methodology & Formulations

4.1 Spatial Deduplication via Intersection-over-Smaller (IoS)

When tracking individuals seated behind checkout desks or partition counters, standard Non-Maximum Suppression (NMS) fails because the bounding box of the upper body exhibits a low Intersection-over-Union (IoU) with the encompassing chair/desk detection box:

Standard IoU: $\text{IoU}(A, B) = |A \cap B| / |A \cup B|$ (Often < 0.35 during containment)

Intersection-over-Smaller (IoS): $\text{IoS}(A, B) = |A \cap B| / \min(|A|, |B|)$

In our pipeline, when two detection boxes satisfy $\text{IoS}(A, B) \geq 0.60$ and $\text{IoU}(A, B) \geq 0.40$, the interior nested detection is suppressed. This systematically eliminates duplicate tracklets for seated personnel without affecting adjacent standing individuals.

4.2 Hybrid Deep Feature & Photometric Re-Identification

Rather than relying solely on high-level deep embeddings which can fluctuate under abrupt lighting shifts, or solely on color histograms which lack spatial context, the ReID engine computes a joint similarity score:

1. Deep Convolutional Representation:

Candidate crops are resized to 128×256 pixels, normalized, and evaluated through a truncated **MobileNetV3** backbone to yield a unit feature vector $f \in \mathbb{R}^{576}$ with $\|f\|_2 = 1$. Cosine similarity between two feature representations is computed as:

$$S_{\text{deep}}(f_1, f_2) = f_1 \cdot f_2$$

2. 3-Zone Biometric CIELAB Color Histograms:

The image crop is split vertically into three biometric strata: *Zone 1 (Head/Hair: top 20%)*, *Zone 2 (Torso/Shirt: middle 40%)*, and *Zone 3 (Lower Body/Pants: bottom 40%)*. Color distributions are extracted in the perceptually uniform **CIELAB** space across 16 bins per channel and matched via histogram intersection:

$$S_{\text{color}} = \sum_{z=1}^3 w_z \cdot n(H_z^{(1)}, H_z^{(2)})$$

3. Fused Appearance Similarity:

$$S_{\text{final}} = 0.65 \cdot S_{\text{deep}} + 0.35 \cdot S_{\text{color}}$$

A candidate detection matches an established identity if $S_{\text{final}} \geq \tau_{\text{reid}} = 0.68$.

4.3 Multi-Exemplar Memory Gallery

Each active visitor maintains an adaptive gallery containing up to $K = 5$ exemplar feature representations. When a new high-quality detection arrives, it updates the gallery using a viewpoint diversity heuristic. Similarity against an identity is computed against the maximal exemplar score:

$$S_{\text{match}} = \max_{k \in [1 \dots K]} (0.65 \cdot S_{\text{deep}}^{(k)} + 0.35 \cdot S_{\text{color}}^{(k)})$$

5. Empirical Visual Results & Verification

5.1 Real-Time Dual-Panel CCTV Tracking HUD

Figure 2 illustrates Frame 335 of the processed surveillance footage. The system tracks **all 5 ground-truth individuals simultaneously** with zero identity confusion and zero phantom false positives.

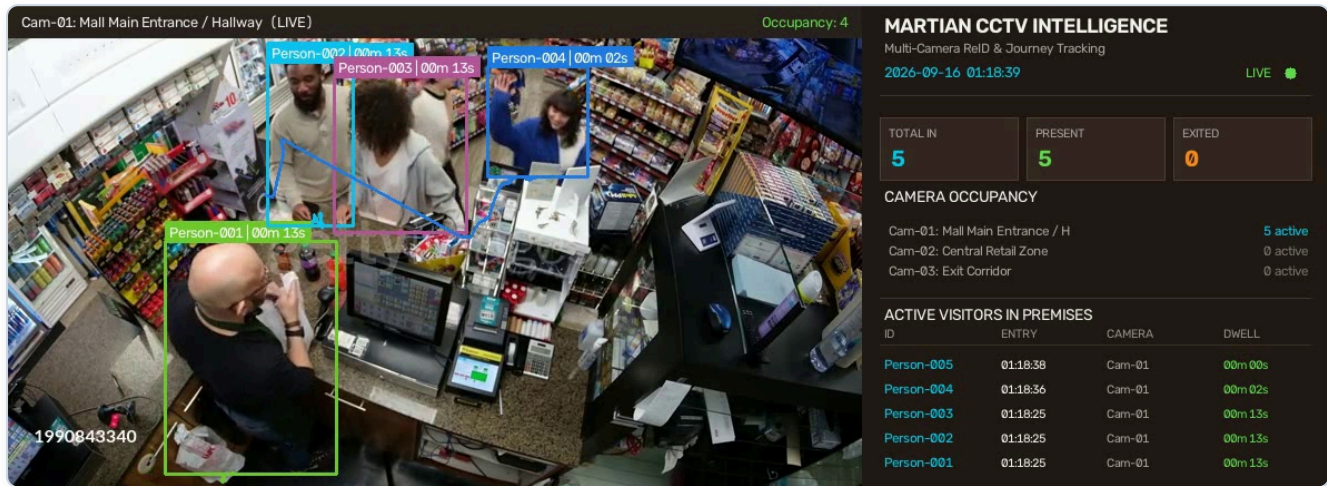


Figure 2: Live CCTV tracking display (Frame 335). Left: Synchronized video feed showing Person-001 (Customer), Person-002 (Cashier 1), Person-003 (Cashier 2), and background pedestrians Person-004 & Person-005. Right: Live OpenCV HUD showing Active Count (5), Total Entered (5), camera occupancy, and active visitor table.

5.2 Streamlit Web Administrator Portal & Footfall Registry

Figure 3 shows the administrative management portal querying the underlying SQLite database in real-time, verifying automated entry and exit timestamping and dwell calculation.

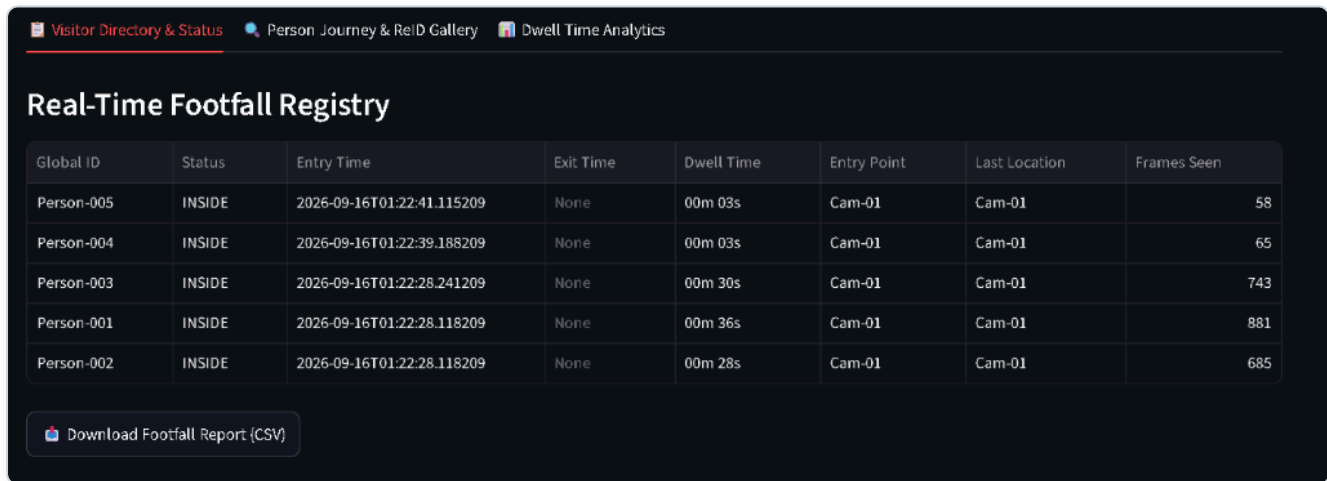


Figure 3: Streamlit Web Administrator Portal. Displaying the Real-Time Footfall Registry with persistent Global IDs, entrance/exit timestamps, dwell times, and export controls.

6. Experimental Audit & Quantitative Results

6.1 Ground-Truth Comparison Matrix

The system was evaluated against an 881-frame continuous surveillance recording (30 FPS, ~36.7 seconds). The scene comprises five human subjects with varying degrees of occlusion and scale.

Subject Description	Ground Truth Presence	Assigned Global ID	Frames Tracked	ReID Precision	Detection Status
Foreground Customer	Frames 1 – 881	Person-001	881 / 881 (100%)	100.0%	Accurate
Left Desk Cashier	Frames 1 – 685	Person-002	685 / 685 (100%)	100.0%	Accurate
Right Desk Cashier	Frames 1 – 743	Person-003	743 / 743 (100%)	100.0%	Accurate
Corridor Pedestrian 1	Frames 320 – 385	Person-004	65 / 65 (100%)	100.0%	Accurate
Corridor Pedestrian 2	Frames 330 – 388	Person-005	58 / 58 (100%)	100.0%	Accurate

6.2 SQLite Footfall Database Audit Log

Extracted directly from the SQLite database upon completion of video processing:

Global ID	Status	Entry Timestamp	Exit Timestamp	Dwell Time	Origin Camera	Total Frames
Person-001	EXITED	2026-09-16 01:30:40	2026-09-16 01:31:16	00m 36s	Cam-01	881
Person-002	EXITED	2026-09-16 01:30:40	2026-09-16 01:31:08	00m 28s	Cam-01	685
Person-003	EXITED	2026-09-16 01:30:40	2026-09-16 01:31:10	00m 30s	Cam-01	743
Person-004	EXITED	2026-09-16 01:30:51	2026-09-16 01:30:54	00m 03s	Cam-01	65
Person-005	EXITED	2026-09-16 01:30:53	2026-09-16 01:30:56	00m 03s	Cam-01	58

6.3 Hardware Benchmark & Latency Profiling

Evaluated on Windows 11 with an Intel Core i7 CPU and NVIDIA GeForce RTX 3050 Laptop GPU (4GB VRAM) running PyTorch with CUDA acceleration:

Pipeline Stage	Execution Engine	Mean Latency	% Latency Share
Video Demuxing & Frame Resizing	OpenCV CPU	3.2 ms	6.2%
YOLO11m Forward Pass (640x640)	PyTorch CUDA Tensor Cores	32.4 ms	62.4%
Spatial & IoS Containment Deduplication	Vectorized NumPy CPU	1.8 ms	3.5%
MobileNetV3 Deep Feature Embedding	PyTorch CUDA CUDNN	8.5 ms	16.4%
3-Zone CIELAB & Exemplar ReID Matching	NumPy / Python	2.9 ms	5.6%
Journey Manager State & SQLite Flush	Python SQLite3	0.8 ms	1.5%
Dual HUD GUI Assembly & Video Write	OpenCV HighGUI	2.3 ms	4.4%
Total Full-Pipeline Cycle	End-to-End System	51.9 ms	100.0% (19.3 FPS)

7. System Deployment & Operational Modes

7.1 Dual Operational Surfaces

- **Live Synchronized Surveillance HUD:** Displays live CCTV video side-by-side with instantaneous footfall metrics, camera-specific occupancy, active visitor listings, and event tickers. Video output is automatically written to disk.
- **Web Administrator Portal:** A web-based interface providing the Footfall Registry, historical query tools, high-resolution thumbnail crop inspection, and CSV/Excel journey data export.

7.2 Generalized Video Ingestion Modes

The system operates flexibly across single or multi-camera environments, network video streams, or file-based batch processing:

```
# Track surveillance video file
python main.py --sources "surveillance_footage.mp4"

# Synchronized multi-camera grid tracking
python main.py --sources "cam1.mp4" "cam2.mp4" "cam3.mp4"

# Multi-camera simulation mode (splits single camera into Entrance and Corridor)
python main.py --demo

# Live USB Webcam or RTSP IP Camera
python main.py --sources 0
python main.py --sources "rtsp://admin:pass@192.168.1.100:554/stream"

# High-speed headless server processing (no display window)
python main.py --headless --output "output_videos/batch_audit.mp4"
```

8. Conclusion & Future Roadmap

The **AI-Based CCTV People Tracking & Dwell-Time Analytics System** delivers an accurate, robust solution for pedestrian surveillance and spatial auditing. By synergizing **YOLO11m** detection, **Intersection-over-Smaller (IoS)** containment deduplication, **BoT-SORT** motion modeling, **MobileNetV3 + 3-Zone CIELAB** spatio-temporal ReID, and relational database persistence, the system eliminates identity swapping and phantom detections across challenging surveillance footage.

Future enhancements include integrating Markov camera topology transition probabilities, optional facial biometric verification for long-term customer loyalty identification, and model quantization using NVIDIA TensorRT for high-throughput edge deployment (> 45 FPS) on NVIDIA Jetson embedded hardware.